

1. Experiment: RFDiffusion3-Based Protein Design**2. Time:** 2025.11.04-2025.11.14**3. Member:** Kaiqing Zhang, Yang Jin, Binxuan Zhang, Qiwen Jiang, Yue Fu**4. Method:**

(1) Structural analysis and hotspot identification

- ① The crystal structure of the IL-6/sIL-6R α /gp130 ternary complex (PDB: 1P9M) was used as the template. For the IL-6 binding site design, the gp130 chain was removed and the IL-6 structure (chain B) was retained as the target, with residues renumbered to match the full-length sequence (Leu19 and Tyr31 corresponding to Leu1 and Tyr13 in the crystal structure, respectively). For the IL-6R α binding site design, the IL-6 and gp130 chains were removed and the IL-6R α structure was retained as the target; Trp249 and Trp296 in the full-length sequence corresponded to Trp154 and Trp201 in the crystal structure, respectively.
- ② Chains were extracted from all structures using PyMOL and residues were renumbered. Key epitope residues were identified as hotspots and used as structural constraints for subsequent design. Literature search and PyMOL analysis of the interleukin-6 family signaling complex interfaces were also performed to exclude potential non-specific binding sites.

(2) Backbone generation using RFDiffusion3

Binder backbones were generated using RFDiffusion3 targeting the Leu19/Tyr31 interface of IL-6 (Part A), the Trp249 interface of IL-6R α (Part B), and the Trp296 interface of IL-6R α (Part C). For each of the three sites, 40 batches were run with 10 structures per batch, generating 400 candidate backbones per site and 1,200 backbones in total.

(3) Core principles of RFDiffusion3

RFDiffusion3 used target-structure and hotspot-residue constraints to generate diverse binder backbones while preserving the target coordinates.

5. Result:

- (1) We aimed to design a *de novo* protein capable of specifically binding the IL-6/sIL-6R α complex. The designed protein comprised three parts: fragment A targeting the IL-6 binding interface, fragment B targeting the Trp249 site of IL-6R α (the gp130-binding site), and fragment C targeting the Trp296 site near the membrane-proximal region of IL-6R α .
- (2) Using the IL-6/sIL-6R α /gp130 ternary complex crystal structure (PDB: 1P9M) as a template, three key epitope regions were identified and hotspot residues were defined as structural constraints. Binder backbones were generated using RFDiffusion3 targeting IL-6 and sIL-6R α , respectively, yielding 40 batches per binding site with 10 structures per batch, totaling 400 candidate backbones.

6. AI Use Statement:

AI tools were used to assist with AlphaFold code generation and debugging. The team reviewed and tested the code. All results are true results. AI-assisted sequence design was performed. Computational predictions were distinguished from experimental validation. Computational prediction generated using AI, not experimentally validated by wet experiment. We employed a combined approach of computational design and experimental verification. Therefore, for results from AI-assisted prediction/simulation, we will continue to conduct wet experimental verification to confirm their validity.